

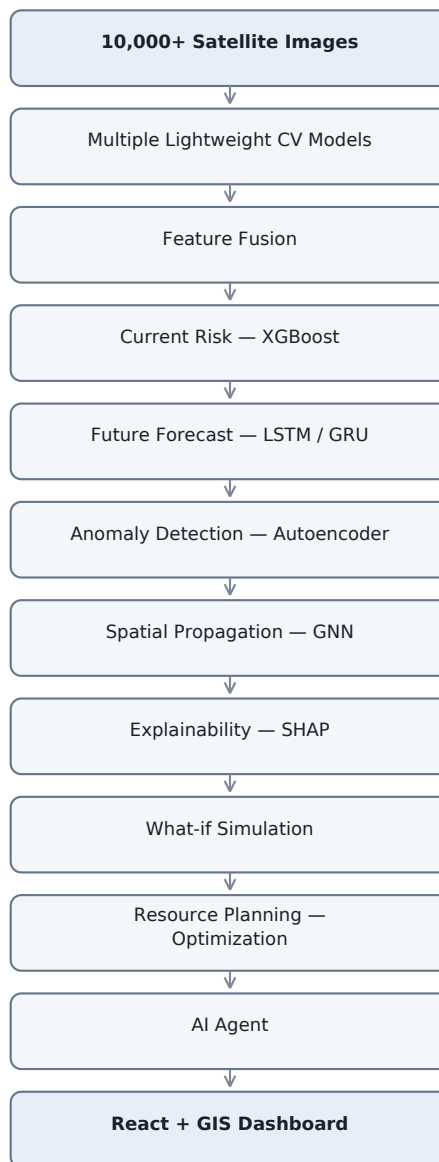
IDMAP-CYCLONE

AI-Powered Dynamic Cyclone Intelligence and Response Planning System for Odisha

COMPLETE PROJECT DOCUMENT — FLOWCHARTS • ARCHITECTURE • PLANS

Scope: Cyclone only • Odisha only • 10,000+ pre-landfall satellite images

Core concept: Multiple lightweight satellite-vision models independently extract cyclone-related visual intelligence. Their outputs are fused with meteorological, geographical, historical, demographic and infrastructure information and passed through an explainable spatiotemporal decision-support pipeline.



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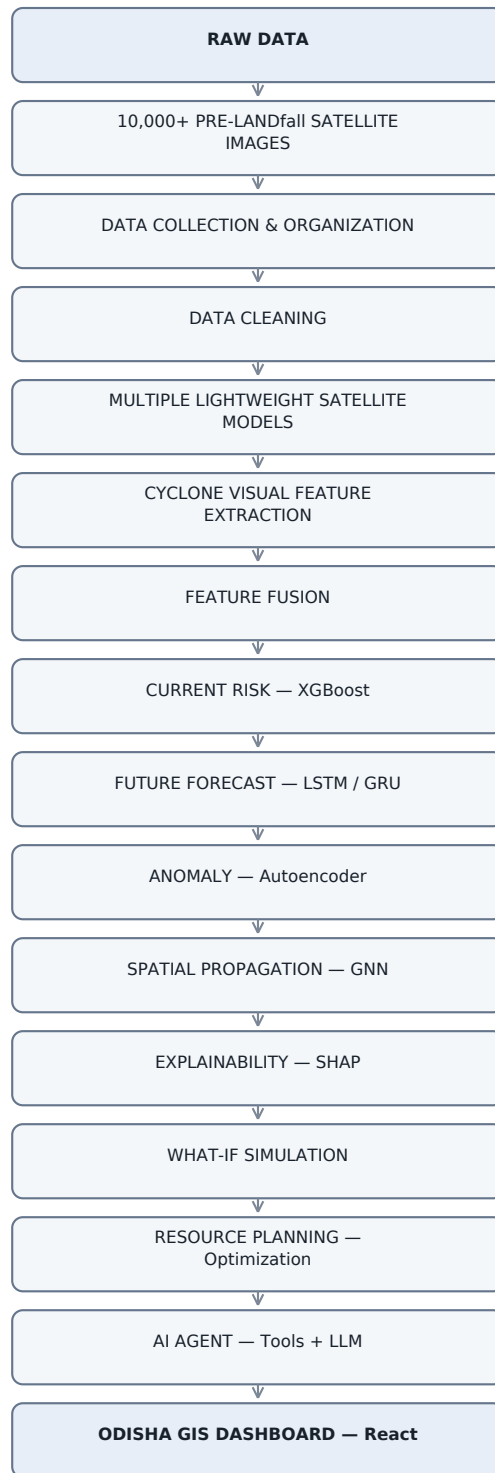
1. PROJECT OVERVIEW AND MENTOR-DIRECTED SCOPE

The project has been narrowed to a single disaster type—cyclone—and a single geographic focus—Odisha. The primary data source is a collection of more than 10,000 satellite images captured before cyclone landfall.

The mentor's computational direction is to avoid a single heavy GPU-dependent training process by dividing the image workload into manageable subsets and training multiple models independently. The outputs of those models are then used as structured inputs to the proposed cyclone-risk and response-planning flow.

The system is therefore not simply an image-classification project; it is an end-to-end cyclone intelligence and decision-support platform.

High-level flow



2. PROBLEM STATEMENT, OBJECTIVES AND SCOPE

Problem statement

Cyclones approaching Odisha can evolve rapidly before landfall. Raw satellite imagery contains valuable visual information, but turning that information into a practical, explainable and spatially actionable risk picture requires multiple stages of analysis. The proposed system addresses this gap by combining satellite-derived features with other contextual data and downstream predictive, spatial and decision-support models.

Main objective

To develop an Odisha-focused cyclone intelligence system that converts pre-landfall satellite observations into current-risk estimates, future forecasts, anomaly signals, spatial risk propagation, explainable predictions, scenario analysis and resource-prioritization recommendations.

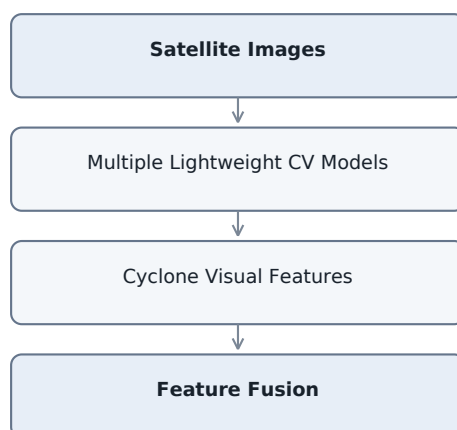
Specific objectives

- Process and organize a 10,000+ image satellite dataset.
- Reduce peak training burden through independently trained lightweight models.
- Extract complementary cyclone-related visual features.
- Fuse satellite-derived features with meteorological and contextual information.
- Predict current regional cyclone risk using XGBoost.
- Forecast future cyclone/risk evolution using LSTM or GRU.
- Detect unusual cyclone evolution using an Autoencoder.
- Model spatial relationships across Odisha using GNN.
- Explain risk predictions using SHAP.
- Support what-if cyclone scenarios.
- Prioritize emergency resources using optimization.
- Provide a tool-using AI assistant.
- Present results through a React + GIS dashboard.

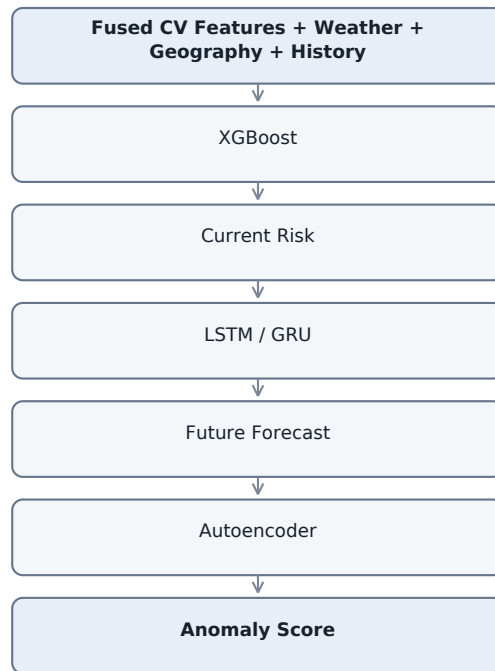
3. SYSTEM ARCHITECTURE

Layer 1 — Computer Vision Intelligence

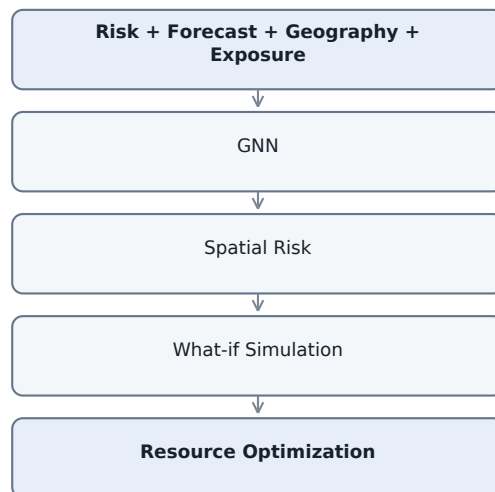
Purpose: understand what the cyclone looks like in satellite imagery.



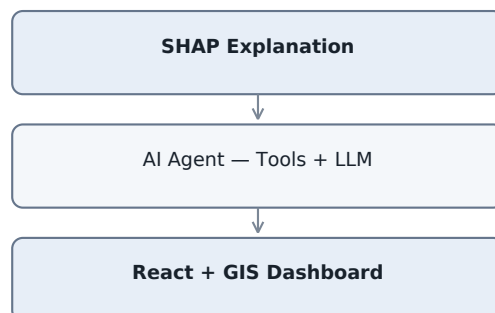
Layer 2 — Predictive Intelligence



Layer 3 — Spatial and Decision Intelligence



Layer 4 — Human Interaction



4. DATASET STUDY AND DATA STRATEGY

This is the first implementation phase. Exact CV tasks and model architectures must be finalized only after examining actual labels, metadata and image characteristics.

Dataset study checklist

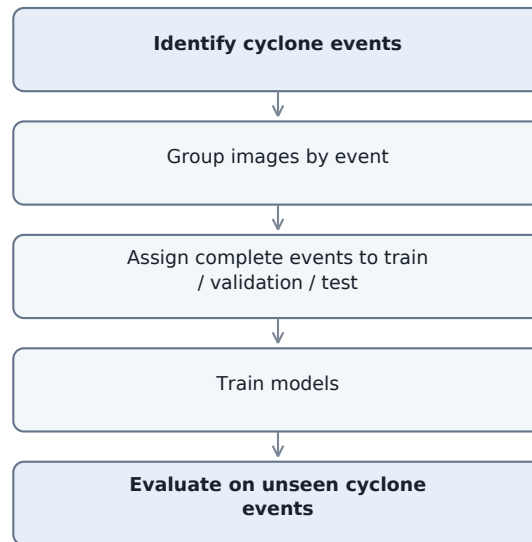
- Total number of images.
- Unique cyclone/event IDs.
- Images per cyclone.
- Image dimensions, channels and file formats.
- Satellite/sensor source if known.
- Timestamps and temporal ordering.
- Confirmation that images are pre-landfall.
- Available labels and annotations.
- Geographic metadata.
- Image quality issues such as blur, missing data or duplicates.
- Class balance and annotation balance.
- Potential data leakage between train/validation/test.

Dataset inventory template

Field	Value / status to determine
Total images	10,000+
Cyclone events	To be identified
Images per event	To be measured
Dimensions	To be measured
Channels	To be inspected
Satellite source	To be identified
Timestamps	To be verified
Pre-landfall status	To be verified
Labels	To be inventoried
Annotations	To be inventoried
Location metadata	To be verified

Event-wise splitting

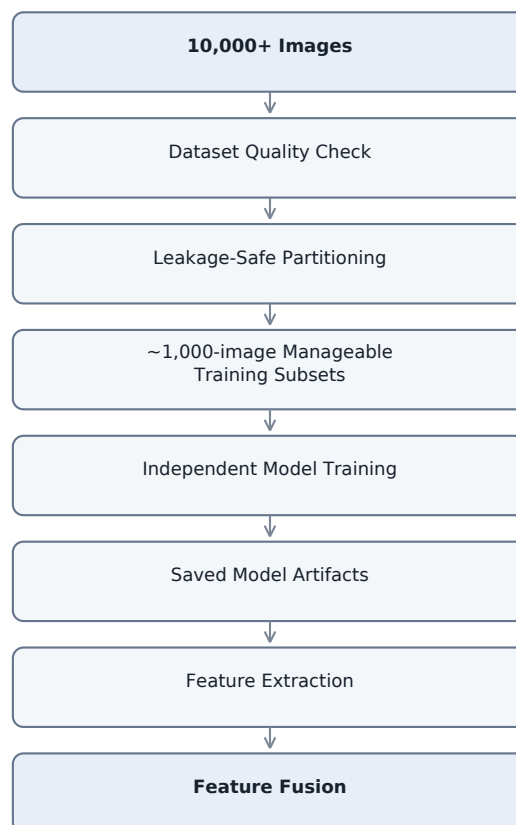
Images from the same cyclone can be highly correlated. Randomly splitting individual images may place nearly identical stages of the same cyclone in both training and testing. Prefer event-wise splitting when feasible.



5. MULTI-MODEL SATELLITE VISION ARCHITECTURE

The 10,000+ images are not intended to become ten arbitrary copies of the same training problem. The recommended interpretation is a collection of lightweight, task-specific models, each producing a complementary feature or prediction.

Partition concept



Candidate model roles

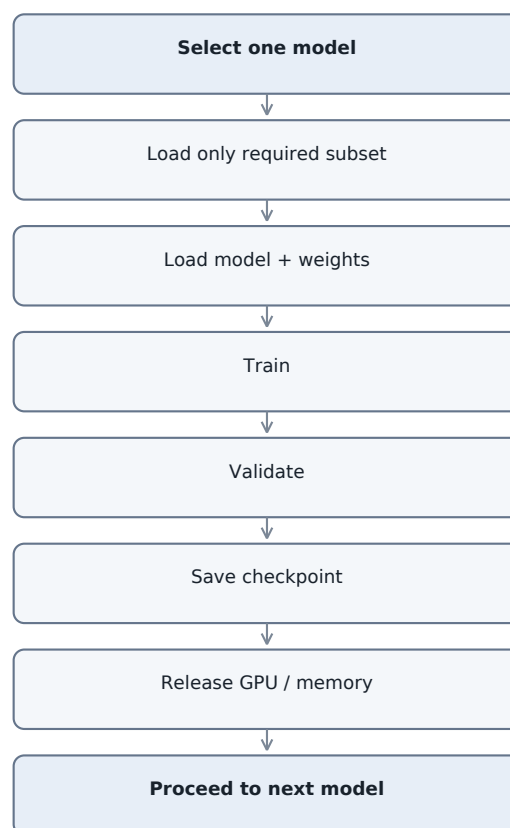
Model	Candidate task	Output used downstream
1	Cyclone/cloud classification	Cyclone probability
2	Intensity-related estimation	Intensity-related feature

Model	Candidate task	Output used downstream
3	Eye detection	Eye confidence/location
4	Cloud structure	Organization/structure score
5	Segmentation	Cyclone/cloud mask features
6	Cloud extent	Spatial extent
7	Temporal/change features	Change score
8	Image anomaly	Satellite anomaly score
9	Spatial/location features	Location-related features
10	Deep feature extraction	Embedding vector

These are candidate roles. The final set must be determined from the actual dataset labels and scientific feasibility.

6. GPU / COMPUTE STRATEGY

The goal is to reduce peak GPU memory and make experiments manageable, not to claim that ten models automatically require less total computation. Training is performed as separate jobs so that all models do not need to occupy GPU memory simultaneously.



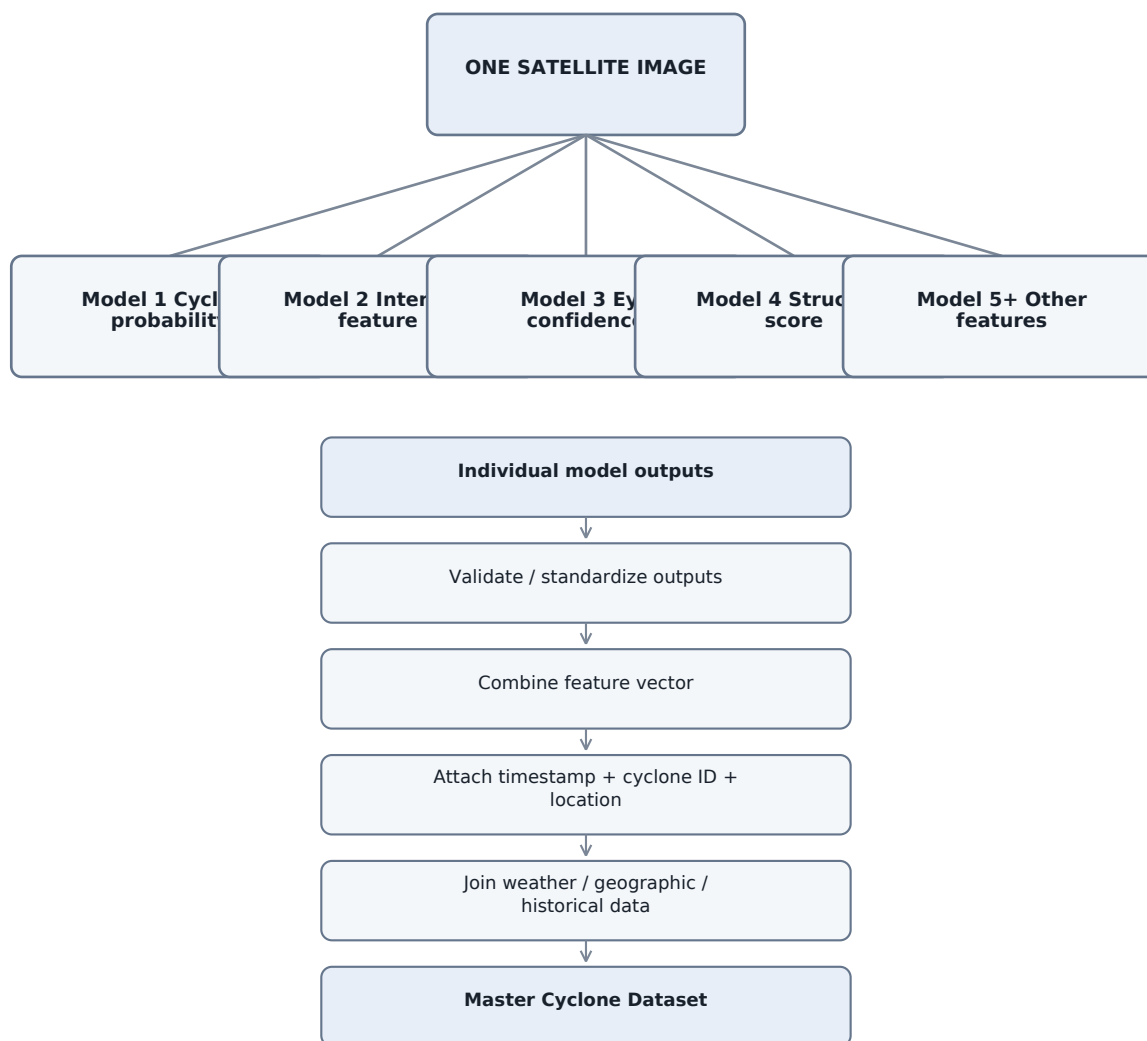
Recommended compute practices

- Transfer learning where appropriate.
- Lightweight architectures where accuracy remains acceptable.
- Freeze backbone layers where practical.
- Small or moderate batch sizes.
- Smaller input resolution when scientifically acceptable.

- Mixed precision when supported.
- Early stopping and checkpointing.
- Sequential model training rather than simultaneous loading.
- Record training time, GPU memory, parameter count and validation performance.

7. FEATURE FUSION AND MASTER DATASET

Feature fusion is the bridge between computer vision and the rest of the IDMAP pipeline.

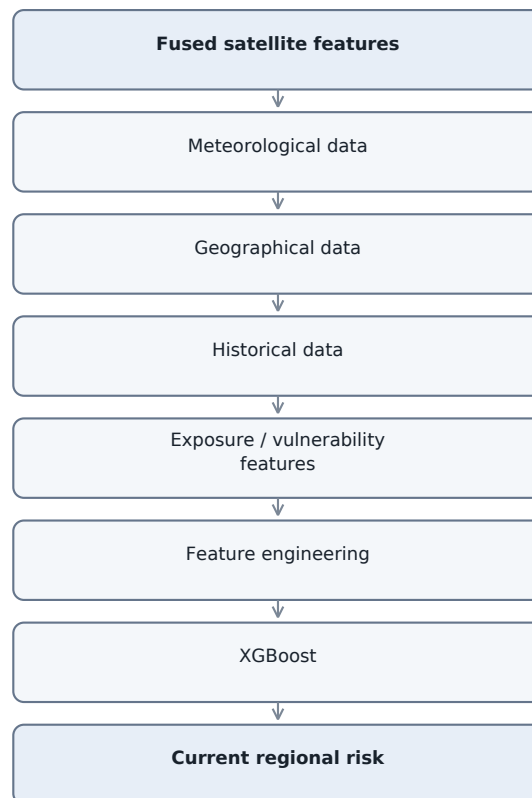


Example master record

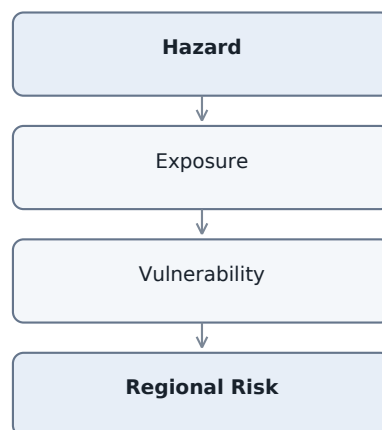
Group	Example fields
Identity	cyclone_id, timestamp
Location	latitude, longitude, district
Satellite	cyclone probability, eye confidence, structure, extent, anomaly
Temporal	change in structure, change in intensity feature
Meteorological	wind, pressure, rainfall, temperature
Geographic	coastal distance, elevation
Exposure	population, infrastructure
Historical	historical cyclone indicators

8. CURRENT RISK — XGBOOST

XGBoost serves as the current-risk prediction model and can combine heterogeneous tabular features produced by satellite models and contextual data.



Risk concept



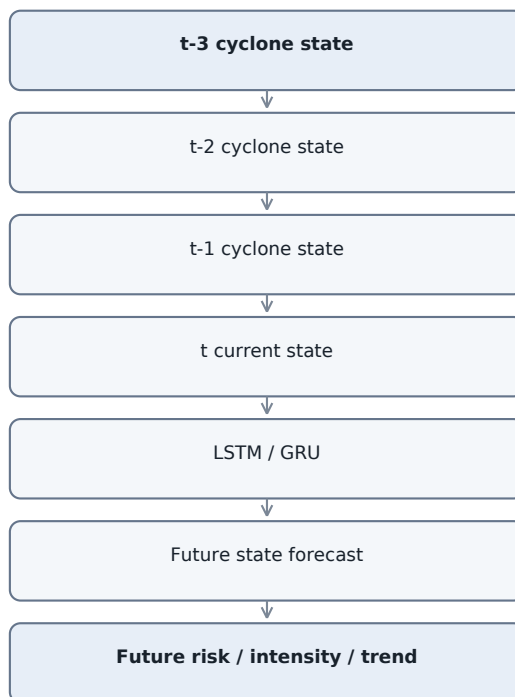
Potential hazard variables include wind intensity, pressure, rainfall and satellite-derived cyclone characteristics. Exposure may include population and infrastructure. Vulnerability may include elevation, coastal proximity, historical impact and infrastructure characteristics.

The output can be a continuous risk score or risk classes such as Low / Moderate / High / Critical, depending on the final target definition.

Evaluation: use classification metrics such as precision, recall, F1, ROC-AUC/PR-AUC or regression metrics such as MAE/RMSE, depending on the target.

9. FUTURE FORECAST — LSTM / GRU

The temporal model receives a sequence of cyclone states rather than isolated images. Each state can contain satellite-derived features, weather variables and current-risk information.

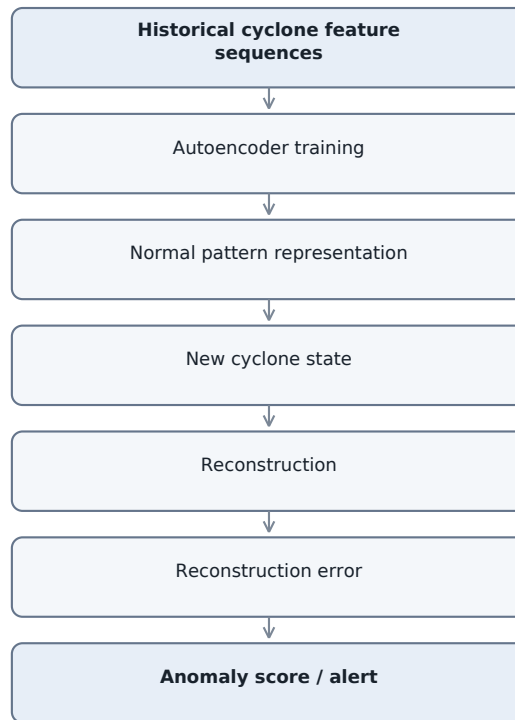


Horizon	Potential output
+6 hours	Risk / intensity-related state
+12 hours	Risk / intensity-related state
+18 hours	Risk / intensity-related state
+24 hours	Risk / intensity-related state

Exact forecast horizons must follow the temporal resolution and historical coverage of the dataset.

10. ANOMALY DETECTION — AUTOENCODER

The Autoencoder learns patterns considered normal in prepared cyclone feature sequences. A high reconstruction error can indicate an unusual observation or evolution pattern.



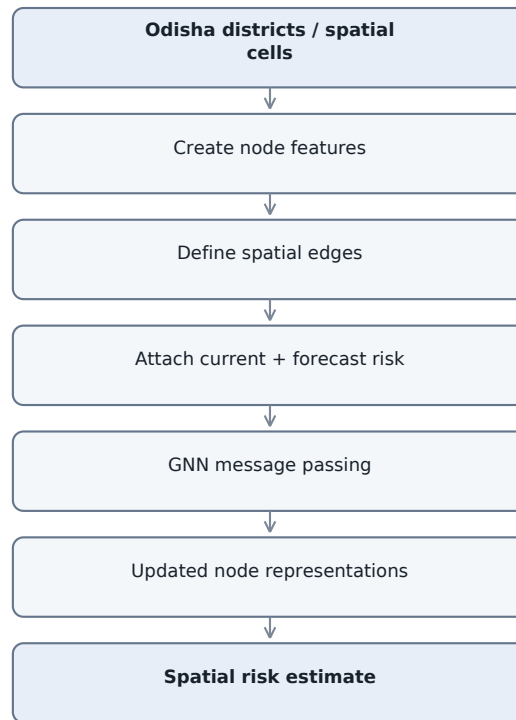
Potential unusual behavior signals

- Rapid increase in wind-related features.
- Rapid pressure decrease.
- Unexpected change in satellite-derived cloud organization.
- Feature combinations rarely observed in training data.

The anomaly threshold must be learned or validated from data rather than chosen arbitrarily.

11. SPATIAL PROPAGATION — GNN

Odisha is represented as a graph. Nodes represent spatial regions, while edges represent meaningful relationships such as proximity or another validated relationship supported by available data.



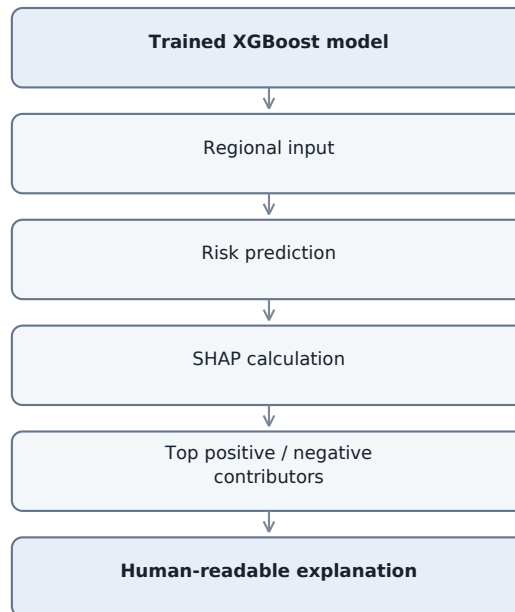
Potential node features

- Current risk.
- Forecast risk.
- Wind / pressure / rainfall.
- Satellite-derived cyclone features.
- Population exposure.
- Elevation and coastal distance.
- Relevant infrastructure information.

If available data cannot support a meaningful graph, retain a simpler spatial baseline instead of forcing a GNN solely because it is a deep-learning component.

12. EXPLAINABILITY — SHAP

SHAP explains the contribution of individual features to a model prediction. This helps the system provide evidence rather than only a black-box risk score.

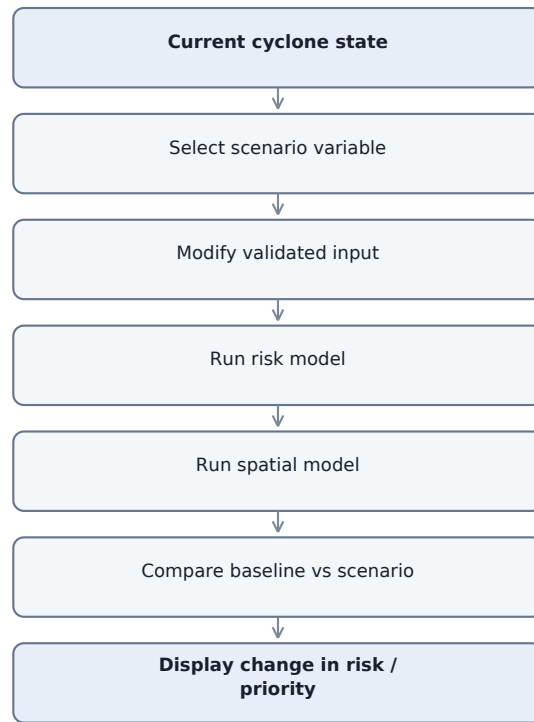


Feature	Illustrative contribution
Satellite structure	Positive
Predicted wind	Positive
Pressure	Positive
Coastal exposure	Positive
Population exposure	Positive
Elevation	Negative / reducing

The table is illustrative; actual SHAP values will come from the trained model.

13. WHAT-IF SIMULATION

The simulation module allows a user to modify selected model inputs and observe how predicted risk and spatial impact change.

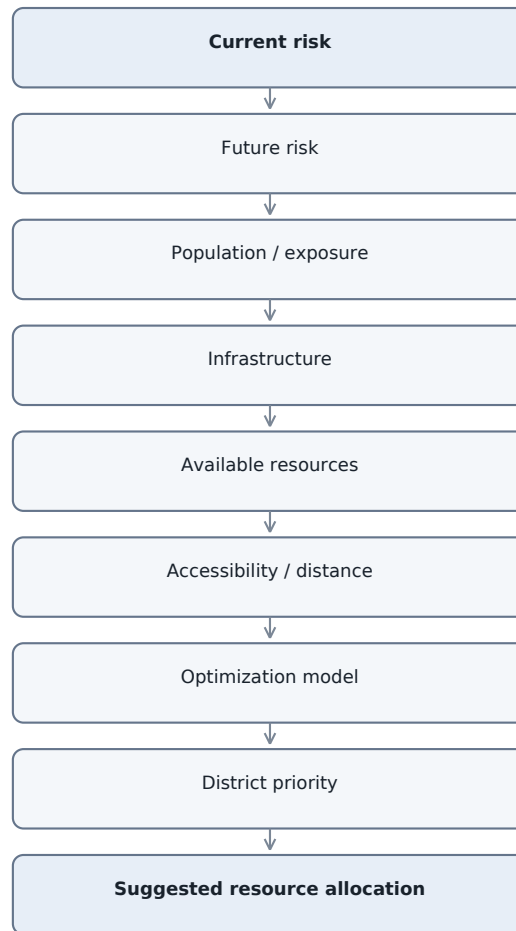


Scenario	Wind	System behavior
Baseline	120 km/h	Calculate current risk
Scenario A	150 km/h	Recalculate risk and spatial impact
Scenario B	Other validated condition	Recalculate and compare

Scenario results should be clearly labeled as model-based simulations, not guaranteed real-world outcomes.

14. RESOURCE PLANNING AND OPTIMIZATION

The resource-planning module converts predicted risk into a prioritization/allocation problem under resource constraints.



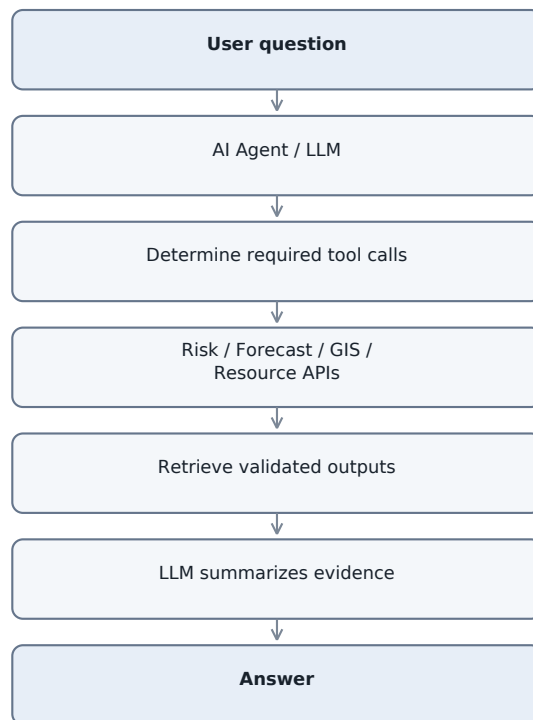
Example resources

- Rescue teams.
- Ambulances.
- Boats.
- Shelter capacity.
- Medical supplies or other project-defined resources.

The system should provide recommendations and prioritization, not autonomous emergency dispatch.

15. AI AGENT — TOOLS + LLM

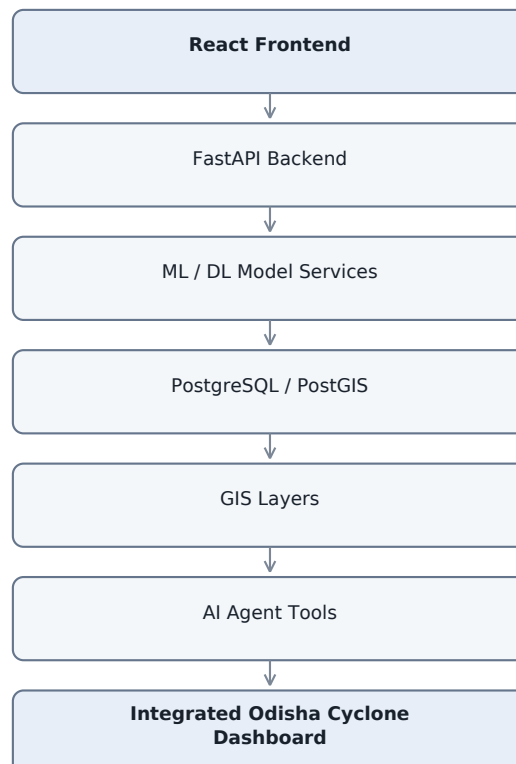
The AI agent is the natural-language interaction layer over the project's actual models and databases. It should use tools to retrieve factual outputs rather than inventing values.



Example: “Which district is most at risk in the next 12 hours?” → retrieve current risk → retrieve forecast → retrieve spatial ranking → retrieve exposure → summarize.

The agent should distinguish model predictions from facts, scenarios and recommendations.

16. DASHBOARD AND GIS ARCHITECTURE

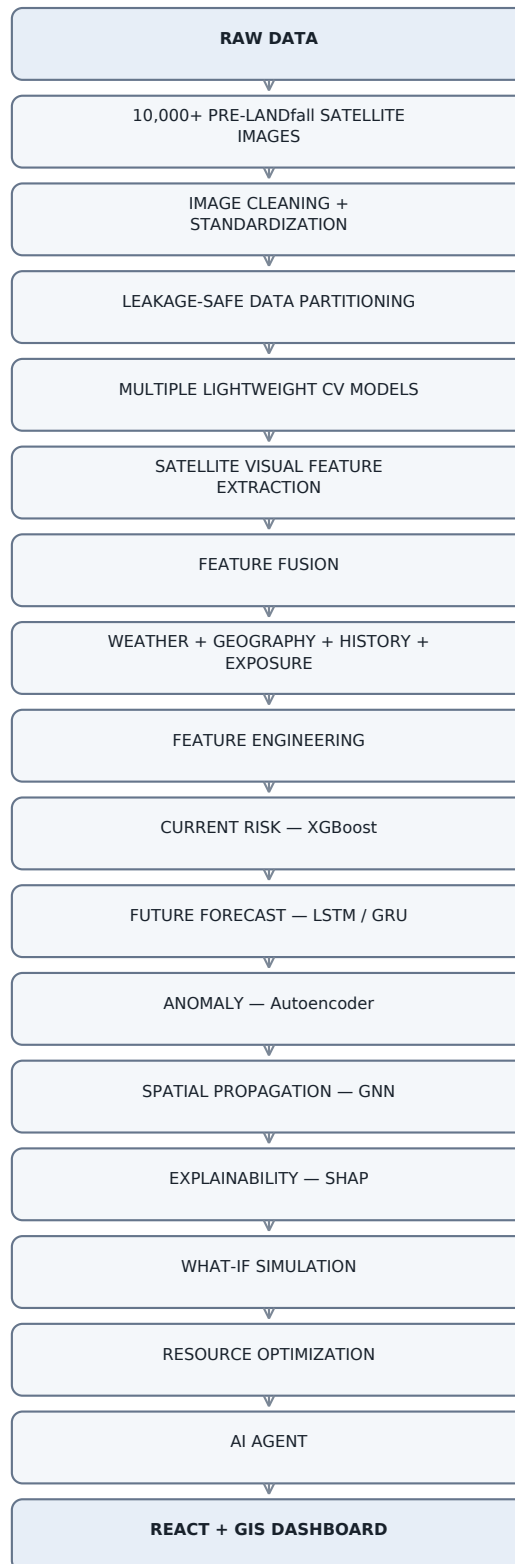


Dashboard modules

- Cyclone overview and current status.
- Interactive Odisha risk map.

- District-level risk ranking.
- Future forecast timeline.
- Anomaly alerts.
- SHAP explanation panel.
- Scenario simulation.
- Resource planning.
- AI assistant.

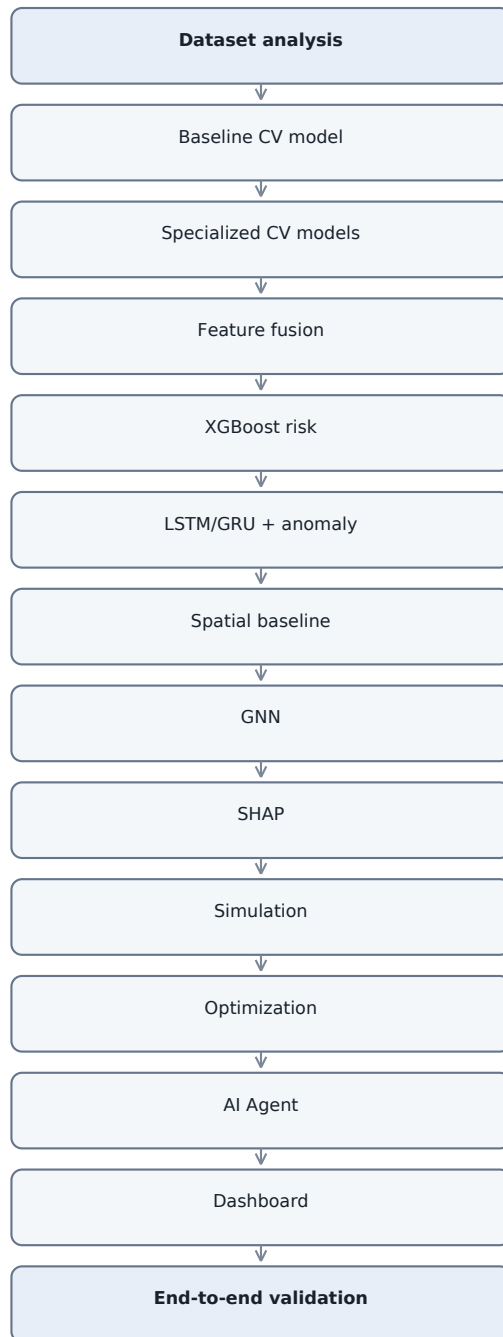
17. COMPLETE END-TO-END DATA FLOW



18. DEVELOPMENT ROADMAP

Phase	Work package	Primary deliverable
1	Dataset study	Dataset inventory, labels, metadata, feasibility
2	Data preparation	Cleaned images and leakage-safe splits
3	CV model development	Independent lightweight satellite models
4	Feature fusion	Structured cyclone intelligence dataset
5	Current risk	Validated XGBoost model
6	Temporal models	LSTM/GRU forecast + Autoencoder anomaly
7	Spatial model	GNN + spatial baseline
8	Explainability	SHAP layer
9	Decision support	What-if + optimization
10	AI agent	Tool-using natural-language assistant
11	Application	React/FastAPI/GIS integration
12	Evaluation	End-to-end validation and ablation
13	Documentation	SRS, report, presentation, demo

Implementation order



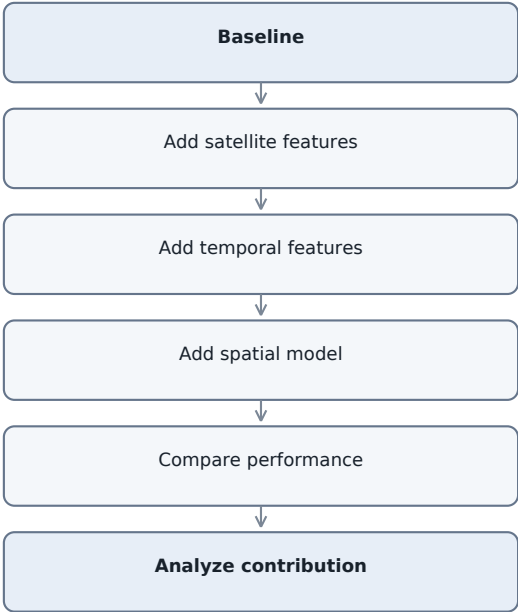
19. MODEL TRAINING AND EVALUATION PLAN

Component	Primary evaluation
Satellite classification	Accuracy, precision, recall, F1, ROC-AUC/PR-AUC
Detection / localization	Task-specific detection metrics if labels exist
Segmentation	IoU / Dice where masks exist
Intensity regression	MAE / RMSE / R^2 as appropriate
XGBoost risk	Classification or regression metrics
LSTM/GRU	MAE / RMSE / temporal forecast error
Autoencoder	Reconstruction error + validated anomaly performance
GNN	Node-level classification/regression metrics
SHAP	Explanation consistency / fidelity checks

Component	Primary evaluation
Simulation	Sensitivity and scenario consistency
Optimization	Objective value, constraints, baseline comparison
System	Latency, reliability, model-serving success

Ablation study

Compare progressively richer information: contextual data only → satellite features → satellite + temporal features → spatial model. This helps demonstrate whether each major component contributes measurable value.



20. RISKS AND MITIGATIONS

Risk	Why it matters	Mitigation
Same cyclone in train/test	Inflated performance	Event-wise split
Arbitrary 1,000-image subsets	Bias / loss of diversity	Event-aware and balanced partitioning
Unclear labels	Invalid targets	Dataset audit first
Class imbalance	Poor minority performance	Weighted loss / sampling / suitable metrics
Overfitting	Poor unseen-cyclone performance	Augmentation, regularization, event-wise validation
Too many models	Complexity without value	Distinct model responsibilities
GNN without valid graph	Artificial reasoning	Validated graph or baseline
LLM hallucination	Incorrect information	Tool-grounded agent
Simulation misuse	False certainty	Clearly label scenarios
Resource recommendation risk	Operational consequences	Human-in-the-loop
GPU constraints	Training failure	Sequential jobs, lightweight models

Design principle

Do not add a model simply to increase the number of AI techniques. Every model must answer a distinct question and contribute a measurable output.

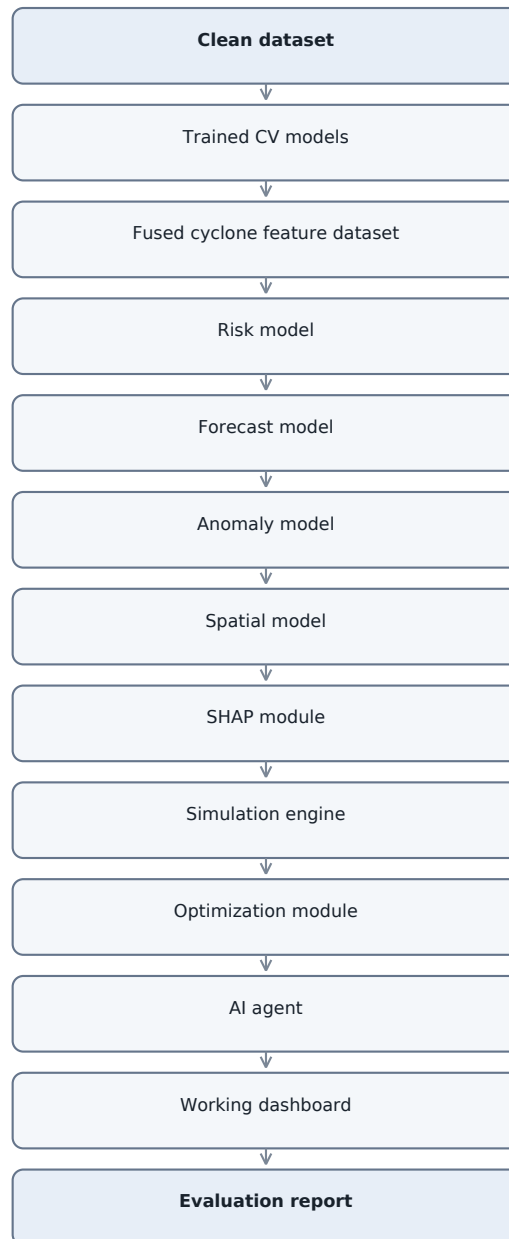
21. ACADEMIC CONTRIBUTION AND EXPECTED OUTCOMES

The project can be positioned as a multi-stage AI decision-support research system rather than a simple cyclone classifier.

Potential academic contributions

- Lightweight multi-model satellite feature extraction.
- Fusion of satellite visual features with meteorological and geographic context.
- Current-risk and temporal-forecast modeling.
- Unusual cyclone-evolution detection.
- Spatial reasoning across Odisha.
- Explainable risk predictions.
- Scenario-based decision support.
- Optimization-based resource prioritization.
- Tool-grounded AI interaction.
- Odisha-focused GIS decision-support application.

Expected outputs



22. IMMEDIATE NEXT STEPS

Before finalizing model-specific implementation, complete this sequence:



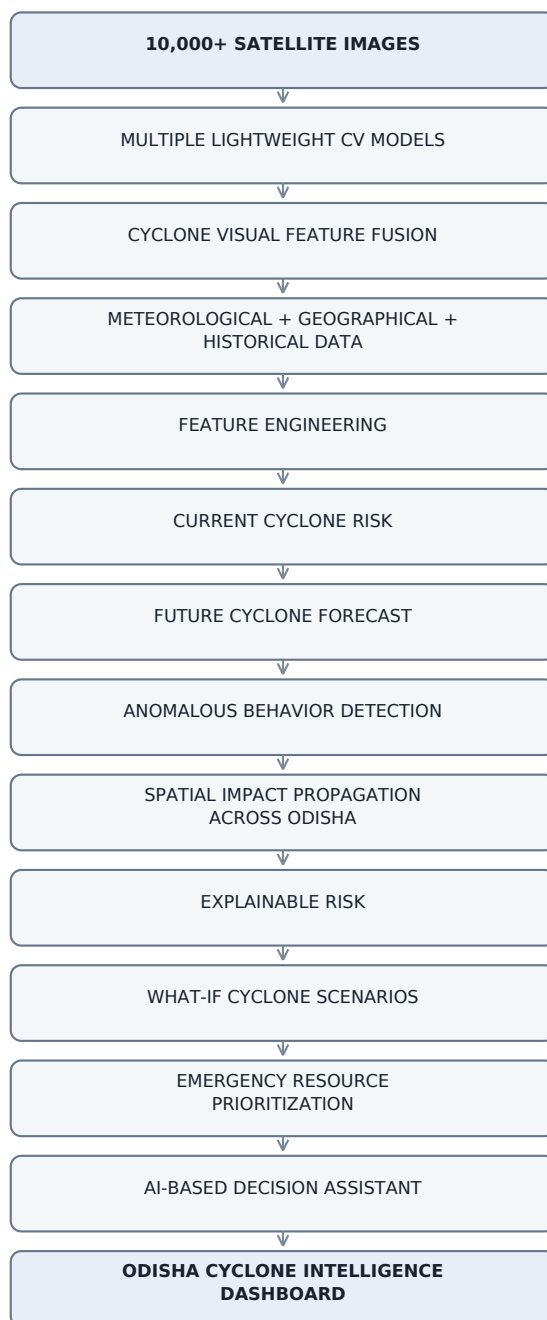
Do not begin with the dashboard. The dashboard should be the final integration layer after the data and model contracts are stable.

23. FINAL PROJECT STATEMENT

IDMAP-Cyclone is an AI-powered dynamic cyclone intelligence and response-planning system designed specifically for Odisha. The system uses a large pre-landfall satellite-image dataset as its primary visual information source. To address computational limitations associated with training on a large image dataset, the images will be organized into manageable subsets and used to train multiple lightweight, task-specific computer-vision models independently. The outputs of these models will be fused into a structured cyclone feature representation. This representation will then be integrated with meteorological, geographical, historical, demographic and infrastructure information and passed through a multi-stage intelligence pipeline consisting of current-risk prediction using XGBoost, future forecasting using LSTM/GRU, anomaly detection using an Autoencoder, spatial risk propagation using GNN, explainability using SHAP, what-if simulation, emergency-resource

optimization and a tool-using AI agent. The final outputs will be presented through an interactive Odisha-focused GIS dashboard.

Final conceptual flow



Key design principle

The project should be presented as an end-to-end decision-support system in which every model has a distinct responsibility. AI is not being added everywhere simply to increase model count; each component is included because it answers a different analytical or operational question.